

Longitudinal Variability of Wearable-Derived Sleep and Heart Rate as a Digital Biomarker for Early Detection of Mild Cognitive Impairment

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Purpose: This study evaluated the potential of wearable-derived longitudinal data for early detection of mild cognitive impairment (MCI). We examined whether variability in nocturnal sleep and heart rate (HR) patterns reflect autonomic nervous system instability associated with cognitive decline.

Materials and Methods: Nocturnal data from 162 participants (111 cognitively normal; 51 MCI) were collected via smart rings over an average of 70 days. To mitigate signal dilution from long-term averaging (LTA), we derived variability-based features (including measures of dispersion, distributional properties, and time-series indices) and compared them to mean-based models. Logistic regression models were constructed through stepwise integration and validated with cross-validation. Key predictors were identified using SHapley Additive exPlanations (SHAP).

Results: Mean-only models showed limited discrimination. In contrast, variability-focused models improved performance, with the model that integrated extended mean, distributional, and time-series features achieving an area under the curve of 0.861 and sensitivity of 0.820. LTA analyses revealed instability of mean effects (signed Cohen's *d* changed in magnitude/direction across 35–70-day windows), supporting a shift away from long-window means. SHAP identified HR_drop variability (short-term variability, mean absolute deviation-based distributional instability, time-bin fluctuation) as top contributors, while sleep-behavior rhythm (e.g., median/trimmed mean of daily sleep count) and circadian-timing features (e.g., mode of sleep midpoint, variability in the timing of the lowest HR) provided complementary information.

Conclusion: Longitudinal wearable monitoring suggests that variability in sleep and HR patterns can capture subtle signs of cognitive decline more effectively than conventional, mean-based measures. These variability-based digital biomarkers may support early MCI screening and personalized intervention strategies.

Key Words: Mild cognitive impairment, digital biomarkers, wearable electronic devices, sleep, heart rate

Received: January 13, 2026 **Revised:** March 4, 2026

Accepted: March 31, 2026 **Published online:** June 8, 2026

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•The authors have no potential conflicts of interest to disclose.

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INTRODUCTION

The potential of wearable devices and smart rings

Mild cognitive impairment (MCI) is a critical prodromal stage of dementia, with an annual conversion rate to Alzheimer's disease (AD) of 10%–15%.¹ Early detection is critical for timely intervention. However, gold-standard diagnostics, such as comprehensive neuropsychological testing and positron emission tomography (PET) imaging, are often invasive, costly, and impractical for continuous longitudinal monitoring.² Wearable devices have emerged as a non-invasive alternative for the

long-term monitoring of sleep, heart rate (HR), and physical activity.^{3,4} While actigraphy and smartwatches provide valuable insights, considerations remain regarding physiological resolution and long-term adherence in certain contexts.^{5,6} In contrast, smart rings—equipped with photoplethysmography (PPG) and triaxial accelerometers—offer superior wearability and stable data collection, particularly during nocturnal sleep.⁷ This study utilizes smart ring-derived longitudinal data to evaluate intra-individual variability as a novel digital biomarker for early MCI detection.

Physiological mechanisms

Sleep and HR are critical indicators of cognitive health.⁵ Reduced slow-wave sleep and rapid eye movement (REM) sleep are hallmarks of early MCI, often linked to impaired glymphatic clearance of amyloid- β and tau proteins.^{8,9} Disruptions of this nature have been shown to cause destabilization of sleep-wake and autonomic nervous system (ANS) regulatory circuits. Notably, these instabilities are often more pronounced in variability metrics than in conventional mean values.¹⁰ Heart rate variability (HRV) serves as an indicator of ANS function, where reduced HRV indicates parasympathetic withdrawal associated with cognitive decline.^{11,12} In MCI, irregular HRV patterns during non-REM sleep highlight the limitations of mean-based analyses and the necessity of incorporating variability measures to capture autonomic instability.¹²

Variability indicators in MCI detection

Emerging evidence suggests that intra-individual variability and circadian rhythm disruption precede mean-level changes in physiological signals.^{13–15} Early cognitive decline often manifests as pattern instability rather than shifts in absolute levels, a phenomenon more sensitively captured by specific variability metrics. Previous studies have identified significant associations between MCI risk and isolated variability indicators, such as dispersion [e.g., standard deviation (SD) or mean absolute deviation (MAD) of nap duration], rhythmicity (e.g., inter-daily stability/intra-daily variability), and distributional characteristics (e.g., kurtosis of HRV).^{13–16} Furthermore, machine learning models utilizing conventional variability measures like standard deviation of NN intervals or root mean square of successive differences (RMSSD) have achieved area under the curve (AUC) values ranging from 0.78 to 0.85.¹¹ However, previous efforts have largely relied on selected individual parameters over mostly short-term periods (14–30 days). There remains a critical lack of research that simultaneously examines long-term nocturnal sleep and HRV through a systematic analytic framework. By integrating comprehensive metrics across dispersion, distribution, and time-series domains, this study aims to capture longitudinal ANS instability more holistically than previous fragmented approaches.

To address these gaps, this study makes several key contributions. First, we utilized longitudinal data spanning approxi-

mately 70 days to capture robust intra-individual variability. Second, we employed high-fidelity monitoring via smart ring-derived signals, ensuring stable and high-compliance ANS quantification. Third, we used linear mixed models (LMM) to assess information loss from mean-based aggregation and applied long-term averaging (LTA) analyses to evaluate signal dilution.¹⁷ Fourth, an integrated analytic framework was established by the stepwise integration of central tendency and multi-dimensional variability metrics. Finally, we identified key digital biomarkers using SHapley Additive exPlanations (SHAP) analysis to ensure clinical interpretability and provide actionable insights for MCI screening.^{18,19}

MATERIALS AND METHODS

Participants

This study utilized a dementia high-risk lifelog dataset from the National Information Society Agency (IRB No. 2-1041055-AB-N-01-2020-50/2020-67). This study utilized a dementia high-risk lifelog dataset from the National Information Society Agency.²⁰ A total of 162 participants [111 cognitively normal (CN); 51 MCI] were selected from an original cohort of 300 individuals aged ≥ 55 years. AD patients were excluded to focus on early-stage detection. Cognitive status was verified via amyloid PET imaging, Mini-Mental State Examination (MMSE),²¹ and the Seoul Neuropsychological Screening Battery.²²

Longitudinal physiological data were collected using a ring-type wearable device (Oura Ring).²³ The device employs PPG sensors to quantify nocturnal HR, HRV, and sleep structure, with prior validation studies demonstrating good agreement with polysomnography.^{23,24} To ensure data robustness, we included participants with ≥ 31 nights of valid data (average: 70 nights). The primary raw variables extracted comprised sleep timing, sleep duration and efficiency, sleep structure (awake/light/deep/REM stages), and nocturnal physiological signals, including HR, HRV, and respiratory rate. All raw wearable variables are presented in Supplementary Table 1 (only online). Daytime activity metrics (e.g., step counts, energy expenditure) were excluded to focus exclusively on nocturnal autonomic instability during sleep. The overall research workflow is illustrated in Fig. 1.

Validation of longitudinal data structure

To investigate whether LTA dilutes clinical signals essential for MCI detection, we performed a four-step validation.

Group comparison

Independent samples t-tests were used to assess if individual daily data points could sufficiently distinguish between CN and MCI groups.

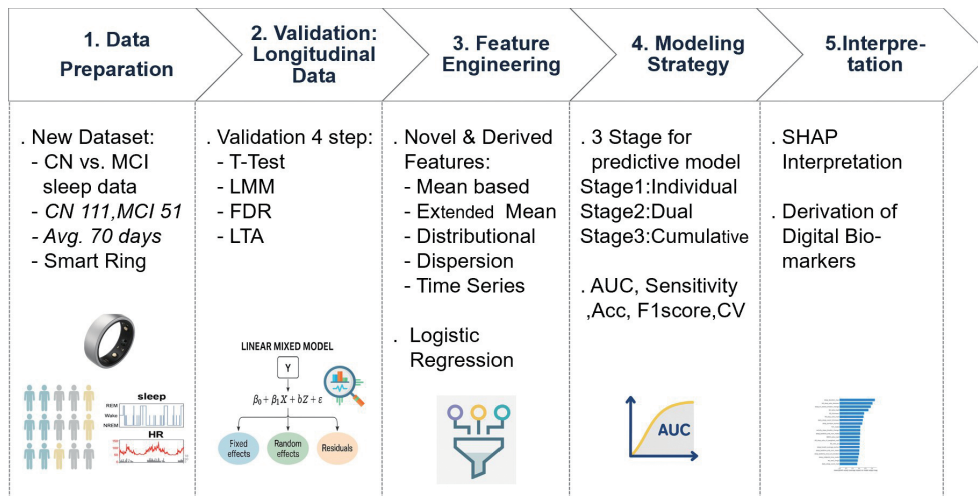


Fig. 1. Overall research workflow. CN, cognitively normal; MCI, mild cognitive impairment; LMM, linear mixed models; FDR, false discovery rate; LTA, long-term averaging; AUC, area under the curve; CV, coefficient of variation; SHAP, SHapley Additive exPlanations.

Hierarchical modeling

LMM accounted for the repeated-measures structure,²⁵ specifying the diagnostic group as a fixed effect and participant IDs as random effects. This allowed for the decomposition of variance into intra- and inter-individual components to quantify information loss during mean-based aggregation.

Multiple comparison correction

False discovery rate (FDR) correction was applied to mitigate type I errors across all physiological variables.

LTA analysis

Daily data were averaged within 35-, 50-, and 70-day windows to obtain one summary value per variable and participant. CN-MCI differences were assessed using Welch's t-tests and signed Cohen's d [95% confidence interval (CI)]. Signal dilution was defined as the instability of mean-based group differences across longer averaging windows and was quantified using the attenuation index ($AI = (d_{35} - d_{70}) / d_{35}$). Based on changes in Cohen's d from 35→70 days, variables were classified into four patterns: attenuation ($AI > 0$), amplification ($AI < 0$), stable ($|d_{35} - d_{70}| < 0.05$), and direction flip (sign reversal of d).

Feature engineering and modeling design

Prior to feature engineering, the dataset—already subjected to standardized quality checks during wearable data collection—underwent additional preprocessing, including removal of missing-timestamp records, correction of infinite values, verification of the absence of NaNs, and z-score standardization of numerical variables. These steps ensured a clean and reproducible dataset for subsequent analysis.

To capture the longitudinal signals of MCI, we engineered multidimensional features and developed predictive models through the following steps.

Feature definition

Daily nocturnal HR, HRV, and sleep structure data were utilized to design novel features in four domains: 1) autonomic and sleep pattern irregularity, 2) sleep rhythm variability, 3) distributional shifts in sleep stages, and 4) sleep regularity.²⁶⁻²⁹

Longitudinal aggregation

Daily metrics were aggregated per participant into five systematic domains, encompassing both central tendency and multidimensional variability:

- Mean-based (M): mean
- Extended mean (EM): median, trimmed mean, mode
- Distributional variability (Dist): minimum, maximum, MAD, kurtosis, range
- Dispersion variability (Disp): SD, coefficient of variation (CV), interquartile range (IQR)
- Time-series variability (TS): time-series-based volatility metrics such as short-term variability, time-bin variability, rolling CV, and moving range (Supplementary Table 2, only online)

Feature selection and modeling

Logistic regression was employed for its robustness and interpretability. Recursive feature elimination identified the optimal predictor subset, excluding variables with high missing rates to ensure clinical reliability.³⁰

Evaluation strategy

Generalizability was assessed via 10-fold cross-validation. Metrics included accuracy, AUC, sensitivity (recall), specificity, and F1-score.

By generating metrics that capture subtle autonomic and behavioral instabilities inherently lost in mean-based approaches, this framework provides a more sensitive methodology for detecting early cognitive decline.

Predictive modeling strategy

We implemented a three-stage experimental framework to identify optimal digital biomarkers while minimizing the information loss associated with simple averaging. The detailed experimental composition is summarized in Table 1.

Stage 1 (individual feature set performance)

The five feature sets—mean-based, extended mean, distributional, dispersion, and time-series variability—were evaluated independently. This stage assessed the unique contribution of each metric group to MCI detection.

Stage 2 (combined feature set performance)

Feature sets demonstrating high discriminative power in Stage 1 were combined to investigate synergistic effects. This step aimed to recover nuanced physiological signals overlooked by single feature sets.

Stage 3 (integrated feature set performance)

Validated feature sets were progressively accumulated to determine the optimal combination for clinical application, ensuring maximum recovery of variability signals.

Model interpretation (SHAP) and derivation of digital biomarkers

To ensure clinical interpretability, we employed SHAP across all experimental stages.³⁰ Based on game-theoretic principles, SHAP quantifies the contribution of each predictor by providing both the direction and magnitude of its impact on MCI classification. This methodology enabled the identification of longitudinal variability indicators that consistently served as robust digital biomarkers. Furthermore, SHAP facilitates personalized interpretation by quantifying feature contributions at the individual participant level. By clarifying how specific indicators, such as sleep fragmentation or HR instability, influence the risk of MCI, SHAP addresses the black-box nature of complex models. This transparency enhances clinical trust-

Table 1. Three-Stage Experimental Design for Modeling and Feature Set Combinations

Stage	Objective	Exp. No.	Composition of feature sets
1	Individual performance comparison	Exp. 1	M (mean-based)
		Exp. 2	EM (extended mean)
		Exp. 3	Dist (distributional variability)
		Exp. 4	Disp (dispersion variability)
		Exp. 5	TS (time-series variability)
2	Pairwise combination with EM	Exp. 6	EM+Dist (Exp. 2+3)
		Exp. 7	EM+Disp (Exp. 2+4)
		Exp. 8	EM+TS (Exp. 2+5)
3	Cumulative integration	Exp. 9	Integration of Exps. 1–4
		Exp. 10	Integration of Exps. 1–5
		Exp. 11	Integration of Exps. 2, 3, 5
		Exp. 12	Integration of Exps. 2–5

Exp., experiment.

worthiness and bridges the gap between machine learning performance and bedside decision-making.³⁰

RESULTS

Baseline characteristics and data feasibility

A total of 11398 nights of longitudinal data were collected, confirming the feasibility of long-term monitoring via smart rings as a viable alternative to one-night polysomnography. As shown in Table 2, there were no significant differences between the CN and MCI groups in gender distribution (53.10% vs. 56.30%) or mean monitoring duration (69.70 days vs. 71.80 days). However, the MCI group exhibited significantly lower MMSE scores (25.81 vs. 27.75, $p < 0.001$), validating the clinical classification of the participants.^{21,22}

Table 2. Demographic and Clinical Characteristics

Variable	CN (n=111)	MCI (n=51)	<i>p</i>
Gender, male (%)	53.10	56.30	>0.050
Mean monitoring duration (days)	69.70	71.80	>0.050
MMSE score	27.75	25.81	<0.001

CN, cognitively normal; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination.

Gender percentages are calculated based on the total study population (excluding Alzheimer's disease cases).

Table 3. Sleep and HR Metrics: Comparison (CN vs. MCI)

Features	T-test (<i>p</i>)	LMM (<i>p</i>)	FDR (<i>p</i>)
N3_ratio	0.006	0.025	0.473
REM_ratio	0.000	0.235	>0.999
NREM_ratio	0.000	0.235	>0.999
SRI	0.640	0.240	>0.999
Sleep_midpoint_time	0.575	>0.999	>0.999
Sleep_duration	0.207	>0.999	>0.999
Sleep_bedtime_end_num	0.000	>0.999	>0.999
TST	0.066	>0.999	>0.999
N1_plus_N2_ratio	0.000	>0.999	>0.999
NREM_proportion	0.000	>0.999	>0.999
HR_drop_ratio	0.000	>0.999	>0.999
WASO	0.000	>0.999	>0.999
SE	0.000	>0.999	>0.999
Sleep_bedtime_start_num	0.000	>0.999	>0.999
Daily_sleep_count	0.000	>0.999	>0.999
Sleep_breath_average	0.000	0.434	>0.999
Sleep_hr_lowest	0.006	0.533	>0.999
Sleep_RMSSD	0.227	0.683	>0.999
Sleep_hr_average	0.000	0.696	>0.999

HR, heart rate; CN, cognitively normal; MCI, mild cognitive impairment; LMM, linear mixed models; FDR, false discovery rate; REM, rapid eye movement; NREM, non-rapid eye movement; SRI, sleep regulatory index; TST, total sleep time; WASO, wake after sleep onset; SE, sleep efficiency.

Comparison of daily and longitudinal data

LMM and FDR-corrected analyses

We assessed whether individual daily measurements could reliably distinguish between groups using LMM and FDR correction (Table 3). While independent t-tests suggested significant differences in sleep efficiency (SE), wake after sleep onset (WASO), and sleep stage ratios ($p < 0.001$),³¹ these findings were nullified when accounting for the hierarchical repeated-measures structure. After LMM decomposition of total variance, p -values increased substantially; following FDR correction, no daily indicators remained statistically significant (N3_ratio $p = 0.473$; all others $p > 0.999$). These results underscore the high noise level in daily raw data and the risk of type I errors.^{25,31}

LTA analysis and signal dilution effects

Across 35-, 50-, and 70-day windows, most variables showed shifts in effect size, indicating that LTA alters CN-MCI separability. As summarized in Table 4, several features attenuated (e.g., sleep HR metrics, sleep duration, N3 ratio), while others amplified (e.g., daily_sleep_count, HR_drop_ratio, stage-ratio features, respiratory measures); a few exhibited direction flips or non-monotonic changes (e.g., WASO, SE). Pattern counts based on AI (35–70) and signed Cohen's d were seven attenuation, nine amplification, two stable, and one direction flip (19 features total), underscoring that mean-only, long-window features are not robust and motivating the use of distributional, dispersion, and time-series variability features.

Performance analysis by feature sets

Stage 1 (individual feature set performance)

To evaluate the unique predictive power of different data characteristics, five statistical feature sets were independently analyzed (Table 5). The mean-based model (Exp. 1) yielded the lowest performance (AUC=0.525), confirming that simple averages are insufficient for capturing subtle MCI-related signals. In contrast, time-series features (Exp. 5) achieved the highest standalone performance (AUC=0.742), followed by distributional features (Exp. 3; AUC=0.701). Extended mean features (Exp. 2) showed moderate efficacy (AUC=0.667), while dispersion-only features (Exp. 4) were comparatively limited (AUC=0.605). These results collectively highlight the limitations of mean-based summaries in isolating clinically relevant MCI signals.

Stage 2 (combined feature set performance)

Since the mean-based model (Exp. 1) performed worst in Stage 1, we excluded simple mean features and combined the better-performing extended mean with other families (Table 5). EM+TS (Exp. 8) yielded the strongest improvement (AUC=0.773, sensitivity=0.680, specificity=0.704).

EM+Dist (Exp. 6) showed modest gains (AUC=0.671), whereas EM+Disp (Exp. 7) demonstrated lower performance (AUC=0.610), suggesting redundancy between dispersion and extended mean features.

Table 4. LTA (35-, 50-, and 70-day) Welch p -Values, Cohen's d , Attenuation Index, and Patterns

Features	p			Cohen's d (MCI-CN)			Attenuation index (35–70)	Patterns
	35 days	50 days	70 days	35 days	50 days	70 days		
Sleep_hr_average	0.464	0.546	0.892	-0.120	-0.107	-0.036	0.700	Attenuation
Sleep_hr_lowest	0.531	0.631	0.906	-0.105	-0.086	0.032	0.694	Attenuation
Sleep_duration	0.504	0.558	0.879	0.111	0.108	0.041	0.631	Attenuation
N3_ratio	0.414	0.302	0.781	-0.142	-0.187	0.079	0.442	Attenuation
N1_plus_N2_ratio	0.088	0.068	0.395	0.305	0.342	0.226	0.259	Attenuation
NREM_proportion	0.080	0.119	0.384	-0.210	-0.202	-0.165	0.211	Attenuation
Sleep_bedtime_start_num	0.008	0.048	0.221	0.359	0.302	0.275	0.234	Attenuation
Sleep_bedtime_end_num	0.022	0.080	0.222	-0.371	-0.315	-0.372	-0.002	Stable
Sleep_midpoint_time	0.679	0.606	0.812	0.067	0.092	0.065	0.026	Stable
WASO	0.060	0.177	0.138	0.323	0.253	-0.322	0.003	Direction-flip
SE	0.061	0.237	0.117	-0.320	-0.211	0.351	-0.097	Amplification
NREM_ratio	0.314	0.398	0.166	0.169	0.149	0.360	-1.134	Amplification
REM_ratio	0.314	0.398	0.166	-0.169	-0.149	-0.360	-1.134	Amplification
Sleep_breath_average	0.396	0.522	0.136	0.135	0.109	0.398	-1.944	Amplification
TST	0.693	0.908	0.462	-0.065	-0.021	0.198	-2.040	Amplification
HR_drop_ratio	0.700	0.572	0.140	-0.063	-0.096	-0.326	-4.212	Amplification
Daily_sleep_count	0.106	0.134	0.011	-0.247	-0.252	-0.640	-1.596	Amplification
Sleep_RMSSD	0.767	0.961	0.598	0.052	0.008	0.141	-1.711	Amplification
SRI	0.960	0.502	0.832	-0.009	-0.123	-0.058	-5.522	Amplification

LTA, long-term averaging; CN, cognitively normal; MCI, mild cognitive impairment; REM, rapid eye movement; NREM, non-rapid eye movement; WASO, wake after sleep onset; SE, sleep efficiency; TST, total sleep time; HR, heart rate; SRI, sleep regularity index.

Table 5. Summary of Model Performance (CV Mean)

Exp. No.	Feature sets	Accuracy	AUC	Sensitivity	Specificity	F1 score
Stage 1: Individual feature set						
Exp. 1	M (mean)	0.533	0.525	0.540	0.530	0.420
Exp. 2	EM (extended mean)	0.625	0.667	0.620	0.627	0.518
Exp. 3	Dist (distributional)	0.623	0.701	0.620	0.626	0.515
Exp. 4	Disp (dispersion)	0.525	0.605	0.560	0.507	0.426
Exp. 5	TS (time-series)	0.683	0.742	0.700	0.675	0.583
Stage 2: Combined feature set						
Exp. 6	EM+Dist (Exp. 2+3)	0.577	0.671	0.600	0.565	0.464
Exp. 7	EM+Disp (Exp. 2+4)	0.565	0.610	0.540	0.577	0.442
Exp. 8	EM+TS (Exp. 2+5)	0.697	0.773	0.680	0.704	0.594
Stage 3: Integrated feature set						
Exp. 9	M+EM+Dist+Disp (Exp. 1–4)	0.747	0.810	0.780	0.731	0.671
Exp. 10	All-feature set (Exp. 1–5)	0.786	0.859	0.800	0.780	0.720
Exp. 11	EM+Dist+TS (Exp. 2, 3, 5)	0.768	0.861	0.820	0.742	0.698
Exp. 12	EM+Dist+Disp+TS (Exp. 2–5)	0.734	0.831	0.720	0.738	0.645

CV, coefficient of variation; AUC, area under the curve.

Stage 3 (integrated feature set performance)

Cumulative integration further improved diagnostic power (Table 5). The all-feature model (Exp. 10) achieved well-balanced performance (AUC=0.859, sensitivity=0.800, specificity=0.780). The variability-focused model excluding the raw mean (Exp. 11; EM+Dist+TS) achieved the highest AUC (0.861) with strong sensitivity (sensitivity=0.820, specificity=0.742). Exp. 12 (EM+Dist+Disp+TS) also performed robustly (AUC=0.831).

Collectively, these results indicate that mean-only features are not robust, whereas combining distributional, dispersion, and time-series variability yields superior discrimination for MCI screening.

Additional analysis

To ensure the robustness of model evaluation and the validity of conclusions; to document that Table 5 metrics were obtained under a consistent baseline setting (minimum ≥35 nights, 7-day variability window with data-leakage safeguards, stratified 10-fold CV); to statistically verify differences across feature families; and to assess methodological suitability via comparisons with nonlinear models, we conducted additional analyses

- **Baseline configuration.** All sensitivity and comparison analyses were performed under the same baseline configuration (minimum ≥35 nights; 7-day variability window) (Supplementary Tables 3 and 4, only online).
- **Between-experiment statistics.** Exp. 10 showed statistically higher AUC than several comparators (BH-adjusted), with no loss in sensitivity at the fixed 0.5 threshold (Supplementary Table 5, only online).
- **Modeling sensitivity.** Using the Exp. 10 feature set, linear models (Logistic/LDA/Elastic-Net) outperformed nonlinear baselines (Random Forest, XGBoost, RBF SVM) (Supplementary Table 6, only online).
- **Feature-to-sample ratio.** With N=162, the FTS ratios were

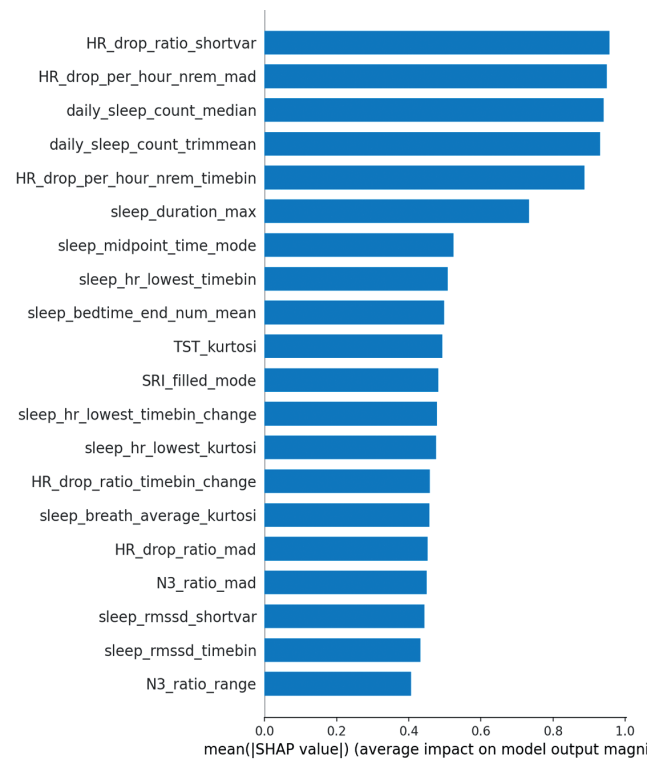


Fig. 2. SHAP result (Exp. 10). SHAP, SHapley Additive exPlanations.

approximately 0.06–0.20 (e.g., Exp. 6≈0.06; Exp. 10–12≈0.20), indicating controlled model complexity relative to sample size.

SHAP-based interpretation

As shown in Fig. 2, the SHAP result (Exp. 10)—which was trained with all feature families included—model predictions were dominated by variability-centered physiological and behavioral features. The most influential predictors arose from heart-rate drop (HR_drop) variability, particularly short-term

variability, MAD-based distributional instability, and time-bin fluctuations, indicating that irregular autonomic responses during sleep are more discriminative for MCI than absolute heart-rate levels. Complementary contributions came from sleep-behavior rhythm indicators (e.g., the median and trimmed mean of daily sleep count) and circadian timing features (e.g., mode of sleep midpoint, variability in the timing of the lowest HR).

Across feature families, time-series variability (e.g., short-term and time-bin variability of HR_drop and RMSSD) and distributional variability (e.g., MAD, kurtosis, maximum, range) showed the highest SHAP values, while mean-based and extended mean features (mean, median, trimmed mean, mode) had smaller effects—indicating that instability and temporal fluctuation—rather than mean-level values—primarily drive MCI classification in Exp. 10.

DISCUSSION

This study demonstrated the clinical relevance of longitudinal variability-based indicators derived from wearable sleep data for early MCI detection.

First, our findings highlight the limitations of mean-based approaches. Traditional mean indicators did not retain statistical significance under hierarchical modeling using LMM with rigorous multiple-comparison control via the FDR.^{23,32} Consistent with this, our LTA analysis demonstrated signal dilution: as aggregation windows expanded from 35 to 70 days, signed Cohen's *d* frequently changed in magnitude—and occasionally in direction—producing AI patterns of seven attenuation, nine amplification, two stable, and one direction flip. These instabilities indicate that simple averaging can obscure or distort subtle pathological fluctuations that matter for early MCI detection. While daily_sleep_count became discriminative only at longer horizons (e.g., 70 days), this was an exception; most mean-level effects were not robust across windows. Collectively, these results argue against exclusive reliance on mean-only, long-window summaries and motivate variability-oriented features—including distributional shifts, dispersion measures, and short-term time-series volatility—to better capture early autonomic and behavioral dysregulation.^{14,17}

Furthermore, variability-oriented features representations outperformed mean-based approaches, as summarized in Table 5, time-series variability (Exp. 5, AUC=0.742) and distributional features (Exp. 3, AUC=0.701) provided the strongest standalone discrimination; EM+TS (Exp. 8) offered the most favorable sensitivity-specificity balance among pairwise combinations; and the variability-focused model excluding the raw mean (Exp. 11: EM+Dist+TS) achieved the highest AUC (0.861) with high sensitivity (0.820) and competitive specificity (0.742). Conceptually, distributional metrics (e.g., MAD, kurtosis) capture non-Gaussian departures in sleep-stage composition and nocturnal physiology, while dispersion metrics (SD, CV,

and IQR) index scale heterogeneity independent of central tendency. Importantly, short-term time-series measures (e.g., rolling CV, moving range, time-bin volatility) are sensitive to fragmented/irregular rhythms, thereby mitigating signal dilution under long-window averaging.^{14,16,17} In parallel, abnormal HR/HRV variability indexes autonomic imbalance and reduced baroreflex/vagal reactivity, weakening brain-heart coupling and promoting microarousals and sleep fragmentation.^{10-13,21} Mechanistically, such ANS lability is consonant with impaired glymphatic clearance (due to disrupted slow-wave dynamics) and neurovascular coupling/microvascular dysregulation, recognizing antecedents of cognitive vulnerability.^{8,33-35}

Overall, a variability-centered framework more faithfully reflects early autonomic and behavioral dysregulation than static means, while offering promising sensitivity-specificity characteristics for future screening applications.

The underlying physiological drivers of these results were further elucidated through SHAP-based interpretation. The SHAP analysis (Fig. 2, Exp. 10) indicates that MCI discrimination is driven less by mean-level values and more by instability, distributional distortion, and temporal fluctuation in physiological and behavioral signals. In particular, HR_drop variability—captured by short-term indices, MAD-based distributional instability, and time-bin fluctuations—ranked highest,^{10-11,21} consistent with blunted parasympathetic bursts, noisy baroreflex gain, and unstable cardiorespiratory coupling during sleep. Complementary contributions arose from sleep-behavior rhythm features (e.g., median/trimmed mean of daily sleep count) and circadian-timing features (e.g., mode of sleep midpoint; variability in the timing of the lowest HR), suggesting ANS desynchrony under circadian misalignment.^{3,14,17} Consistently, distributional (e.g., MAD, kurtosis) and time-series features (e.g., short-term variability, time-bin change) outweighed mean-based/extended mean measures, implying that autonomic lability and temporal dysregulation—rather than static level differences—carry the primary signal of early cognitive vulnerability.²⁴

This view is reinforced by Fig. 3 (HR_drop_ratio comparison). Whereas mean values show minimal group separation, MAD and short-term variability reveal marked irregularity and larger amplitude in the MCI group.^{10,11,13} Crucially, cumulative MAD demonstrates that groups are indistinguishable early on but diverge over time, with MCI participants exhibiting a steeper decline and earlier stabilization of variability—an empirical pattern consistent with irregular autonomic responses, progressive ANS dysfunction, and reduced physiological adaptability with increased rigidity.^{10,11,13,33-35} Together, Figs. 2 and 3 support the view that variability-centered, distribution-aware, and temporally dynamic markers are sensitive, interpretable digital biomarker candidates for the early identification and monitoring of cognitive decline.^{3,14}

Despite these promising results, several limitations of this study should be acknowledged. First, due to data-protection

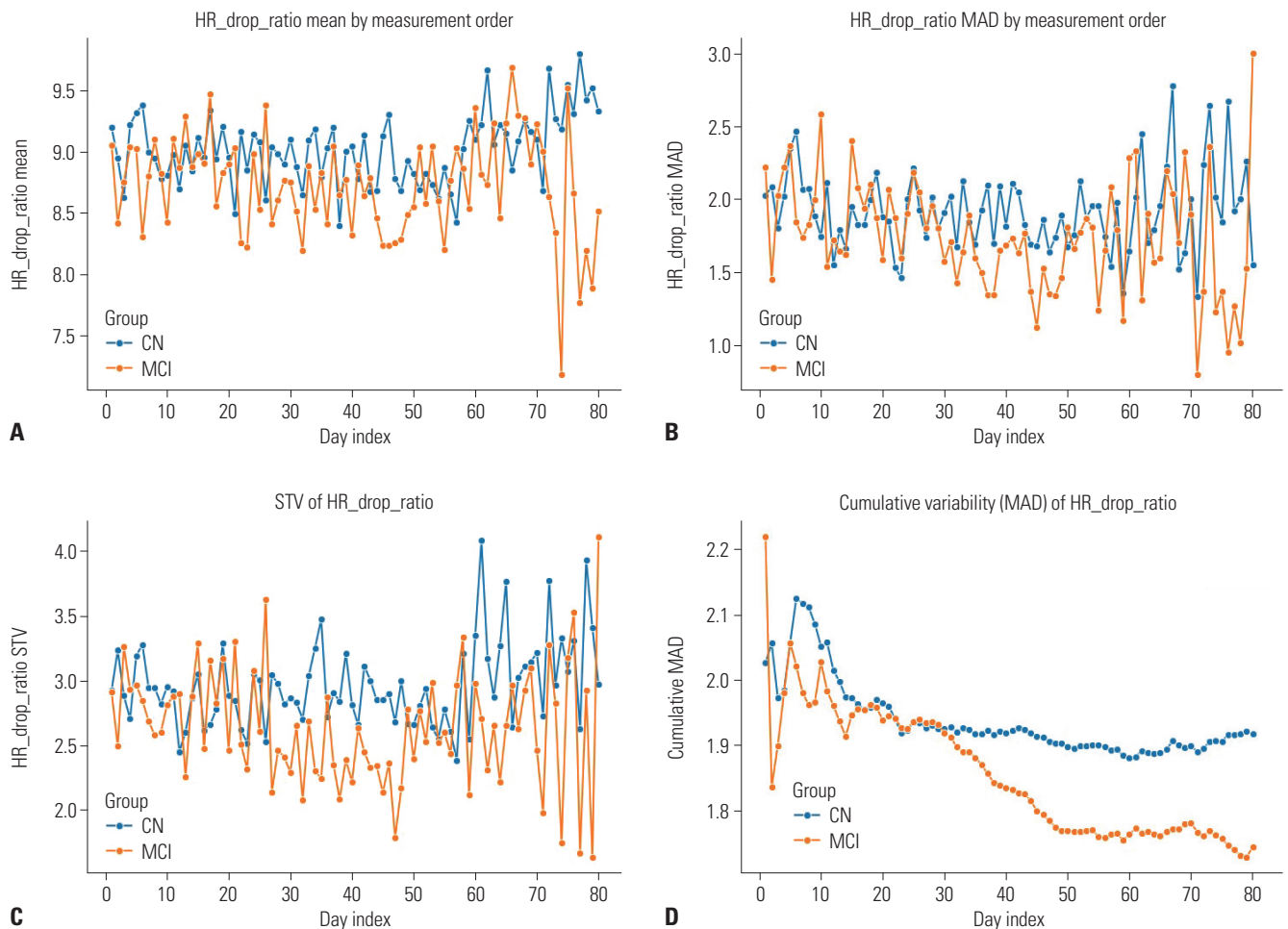


Fig. 3. Comparison of HR_drop_ratio metrics: mean (A), MAD (B), STV (C), and cumulative MAD (D; reflecting distributional variability over time). HR, heart rate; CN, cognitively normal; MCI, mild cognitive impairment; MAD, mean absolute deviation; STV, short-term variability.

constraints, individual-level demographic variables were not accessible, which limited the scope of demographic adjustment analyses. Second, the single-center design and relatively small sample size produced wide CIs in bootstrap validation; although the lower bounds remained clinically meaningful, performance showed sensitivity to resampling. Third, potential sensor noise inherent to wearable devices may have influenced signal quality. We implemented strict anti-leakage procedures and a standardized preprocessing pipeline; nonetheless, some residual bias may remain.

Future research will pursue multicenter validation for broader generalizability. Currently, an independent clinical trial is underway to standardize algorithms and enhance digital biomarker robustness, aiming to bridge the gap between retrospective discovery and practical MCI screening.

In conclusion, this study empirically demonstrates the structural limitations of mean-based approaches in longitudinal wearable data and establishes the superiority of distributional and time-series variability indicators. By identifying key markers of instability in sleep-wake rhythms, autonomic dysregulation (e.g., HR/HRV short-term and time-bin variabil-

ity), and fragmentation of deep-sleep architecture, we provide a methodological foundation for next-generation, wearable-based cognitive health monitoring. Shifting the analytic paradigm from static averages to dynamic instability yields a more sensitive and clinically relevant signature for the early detection of cognitive decline.

AVAILABILITY OF DATA AND MATERIAL

This research used datasets from “The Open AI Dataset Project (AI-Hub, S. Korea).” All data information can be accessed through “AI-Hub (www.aihub.or.kr).”

ACKNOWLEDGEMENTS

This research was supported by the ASTRA Project through the National Research Foundation (NRF) funded by the Ministry of Science and ICT (No. RS-2024-00470056).

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REFERENCES

- Petersen RC, Smith GE, Waring SC, Ivnik RJ, Tangalos EG, Kokmen E. Mild cognitive impairment: clinical characterization and outcome. *Arch Neurol* 1999;56:303-8.
- Weiner MW, Veitch DP, Aisen PS, Beckett LA, Cairns NJ, Green RC, et al. The Alzheimer's Disease Neuroimaging Initiative: a review of papers published since its inception. *Alzheimers Dement* 2012;8(1 Suppl):S1-68.
- Bonvino A, Cornacchia E, Scaramuzzi GE, Gasparre D, Manippa V, Rivolta D, et al. Digital tools for mild cognitive impairment: a systematic review and meta-analysis of diagnostic accuracy and methodological challenges. *Neuropsychol Rev* 2025 Oct 22 [Epub]. Available at: <https://doi.org/10.1007/s11065-025-09680-8>.
- Li A, Li J, Zhang D, Wu W, Zhao J, Qiang Y. Synergy through integration of digital cognitive tests and wearable devices for mild cognitive impairment screening. *Front Hum Neurosci* 2023;17:1183457.
- Ancoli-Israel S, Cole R, Alessi C, Chambers M, Moorcroft W, Polak CP. The role of actigraphy in the study of sleep and circadian rhythms. *Sleep* 2003;26:342-92.
- de Zambotti M, Goldstein C, Cook J, Menghini L, Altini M, Cheng P, et al. State of the science and recommendations for using wearable technology in sleep and circadian research. *Sleep* 2024;47:zsad325.
- Marino M, Li Y, Rueschman MN, Winkelman JW, Ellenbogen JM, Solet JM, et al. Measuring sleep: accuracy, sensitivity, and specificity of wrist actigraphy compared to polysomnography. *Sleep* 2013;36:1747-55.
- Chong PLH, Garic D, Shen MD, Lundgaard I, Schwichtenberg AJ. Sleep, cerebrospinal fluid, and the glymphatic system: a systematic review. *Sleep Med Rev* 2022;61:101572.
- Xie L, Kang H, Xu Q, Chen MJ, Liao Y, Thiyagarajan M, et al. Sleep drives metabolite clearance from the adult brain. *Science* 2013;342:373-7.
- Kong SDX, Hoyos CM, Phillips CL, McKinnon AC, Lin P, Duffy SL, et al. Altered heart rate variability during sleep in mild cognitive impairment. *Sleep* 2021;44:zsaa232.
- Rykov YG, Patterson MD, Gangwar BA, Jabar SB, Leonardo J, Ng KP, et al. Predicting cognitive scores from wearable-based digital physiological features using machine learning: data from a clinical trial in mild cognitive impairment. *BMC Med* 2024;22:36.
- Grässler B, Dordevic M, Herold F, Darius S, Langhans C, Halfpaap N, et al. Relationship between resting state heart rate variability and sleep quality in older adults with mild cognitive impairment. *Int J Environ Res Public Health* 2021;18:13321.
- Phillips AJK, Clerx WM, O'Brien CS, Sano A, Barger LK, Picard RW, et al. Irregular sleep/wake patterns are associated with poorer academic performance and delayed circadian and sleep/wake timing. *Sci Rep* 2017;7:3216.
- Costa M, Goldberger AL, Peng CK. Multiscale entropy analysis of complex physiologic time series. *Phys Rev Lett* 2002;89:068102.
- Ivanov PCh, Hu K, Hilton MF, Shea SA, Stanley HE. Endogenous circadian rhythm in human motor activity uncoupled from circadian influences on cardiac dynamics. *Proc Natl Acad Sci U S A* 2007;104:20702-7.
- Shaffer F, Ginsberg JP. An overview of heart rate variability metrics and norms. *Front Public Health* 2017;5:258.
- Laird NM, Ware JH. Random-effects models for longitudinal data. *Biometrics* 1982;38:963-74.
- Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, et al. From local explanations to global understanding with explainable AI for trees. *Nat Mach Intell* 2020;2:56-67.
- Kelly CJ, Karthikesalingam A, Suleyman M, Corrado G, King D. Key challenges for delivering clinical impact with artificial intelligence. *BMC Med* 2019;17:195.
- AI-Hub. Dementia high-risk group lifelog dataset [accessed on 2026 April 13]. Available at: <https://aihub.or.kr/aihubdata/data/view.do?dataSetSn=226>.
- Folstein MF, Folstein SE, McHugh PR. "Mini-mental state": a practical method for grading the cognitive state of patients for the clinician. *J Psychiatr Res* 1975;12:189-98.
- Ahn HJ, Chin J, Park A, Lee BH, Suh MK, Seo SW, et al. Seoul Neuropsychological Screening Battery-dementia version (SNSB-D): a useful tool for assessing and monitoring cognitive impairments in dementia patients. *J Korean Med Sci* 2010;25:1071-6.
- Svensson T, Madhawa K, Nt H, Chung UI, Svensson AK. Validity and reliability of the Oura Ring Generation 3 (Gen3) with Oura sleep staging algorithm 2.0 (OSSA 2.0) when compared to multi-night ambulatory polysomnography: a validation study of 96 participants and 421,045 epochs. *Sleep Med* 2024;115:251-63.
- Wang R, Blackburn G, Desai M, Phelan D, Gillinov L, Houghtaling P, et al. Accuracy of wrist-worn heart rate monitors. *JAMA Cardiol* 2017;2:104-6.
- Lim AS, Kowgier M, Yu L, Buchman AS, Bennett DA. Sleep fragmentation and the risk of incident Alzheimer's disease and cognitive decline in older persons. *Sleep* 2013;36:1027-32.
- Mander BA, Winer JR, Walker MP. Sleep and human aging. *Neuron* 2017;94:19-36.
- Landry GJ, Best JR, Liu-Ambrose T. Measuring sleep quality in older adults: a comparison using subjective and objective methods. *Front Aging Neurosci* 2015;7:166.
- Duncombe J, Kitamura A, Hase Y, Ihara M, Kalaria RN, Horsburgh K. Chronic cerebral hypoperfusion: a key mechanism leading to vascular cognitive impairment and dementia. Closing the translational gap between rodent models and human vascular cognitive impairment and dementia. *Clin Sci (Lond)* 2017;131:2451-68.
- Iadecola C. The neurovascular unit coming of age: a journey through neurovascular coupling in health and disease. *Neuron* 2017;96:17-42.
- Willie CK, Tzeng YC, Fisher JA, Ainslie PN. Integrative regulation of human brain blood flow. *J Physiol* 2014;592:841-59.
- Gorelick PB, Scuteri A, Black SE, Decarli C, Greenberg SM, Iadecola C, et al. Vascular contributions to cognitive impairment and dementia: a statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2011;42:2672-713.
- Nicolini P, Lucchi T, Abbate C, Inglese S, Tomasini E, Mari D, et al. Autonomic function predicts cognitive decline in mild cognitive impairment: evidence from power spectral analysis of heart rate

- variability in a longitudinal study. *Front Aging Neurosci* 2022;14:886023.
33. Nicolini P, Malfatto G, Lucchi T. Heart rate variability and cognition: a narrative systematic review of longitudinal studies. *J Clin Med* 2024;13:280.
34. Ma J, Chen M, Liu GH, Gao M, Chen NH, Toh CH, et al. Effects of sleep on the glymphatic functioning and multimodal human brain network affecting memory in older adults. *Mol Psychiatry* 2025;30:1717-1729.
35. Cejudo A, Arrojo M, Martín C, Almeida A. AI and wearables for early detection of cognitive impairment and dementia: systematic review. *J Med Internet Res* 2026;28:e86262.